et 8A 5.24 5.29 5.32 5.47 5.51 5.53

5.24

解: $-KX = Ma$

$\Rightarrow \frac{d^2x}{dt^2} + \frac{K}{M}x = 0$ $\Rightarrow \begin{cases} \omega = \sqrt{\frac{K}{M}} \\ \omega' = \sqrt{\frac{K}{M+m}} \end{cases} \Rightarrow T' = \sqrt{\frac{M+m}{M}}\, T$

$\omega'^2 = \frac{M}{M+m}\omega^2$

$x^2 + \frac{V'^2}{\omega'^2} = A'^2$

$x = A\sin\omega t_1$ $\Rightarrow A' = A\sqrt{\frac{m\cos^2\omega t_1}{M+m} + \sin^2\omega t_1}$

$(M+m)V' = mV$, $V = \omega A\cos\omega t_1$

5.29

解 $\begin{cases} mgR - KR = maR + J\beta \\ J = \frac{1}{2}MR^2 \\ \beta R = a \end{cases}$ $\Rightarrow (m+\frac{1}{2}M)a + K(x - \frac{mg}{K}) = 0$

$\Rightarrow \omega = \sqrt{\frac{K}{m+\frac{1}{2}M}}$

$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m+\frac{1}{2}M}{K}}$

5.32 (a) $-k_1x - k_2x = ma$

$\frac{d^2x}{dt} + \frac{k_1+k_2}{m}x = 0$ $\Rightarrow \omega = \sqrt{\frac{k_1+k_2}{m}}$

$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k_1+k_2}{m}}$

(b) $-k_1x - k_2x = ma$

$\frac{d^2x}{dt} + \frac{k_1+k_2}{m}x = 0$ $\Rightarrow \omega = \sqrt{\frac{k_1+k_2}{m}}$

$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k_1+k_2}{m}}$

(c) $K = \frac{k_1k_2}{k_1+k_2}$

$-KX = ma$

$\Rightarrow \frac{d^2x}{dt^2} + \frac{k_1k_2}{m(k_1+k_2)}x = 0$ $\Rightarrow \omega = \sqrt{\frac{k_1k_2}{m(k_1+k_2)}}$

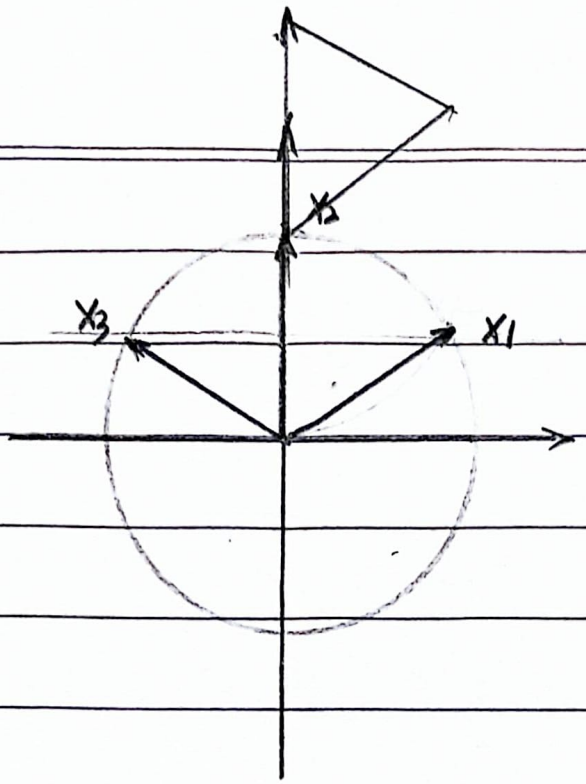
$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k_1k_2}{m(k_1+k_2)}}$

5.47 (1) $k_2 = \frac{2}{6}$ 时振幅最大 $A = 0.6m$

(2) $k_2 = \frac{2}{6}$ 2时振幅最小 $A = 0.2m$

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5.51

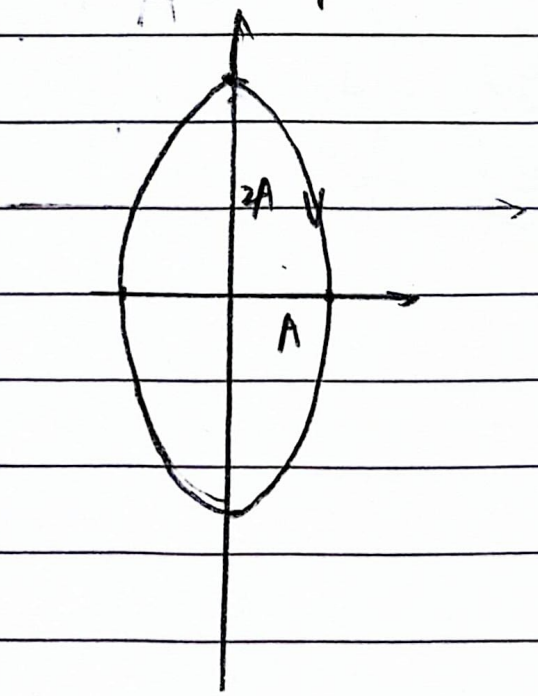


$x = 0.08\cos(120\pi t + \frac{2}{3})m$

5.53 $x = A\cos(\omega t - \frac{\pi}{2}) = A\sin\omega t$

$y = 2A\cos\omega t$

$\Rightarrow x^2 + \frac{y^2}{4} = A^2$



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作业8B 6.9 6.12 6.21 6.27

作业8C 6.34 6.36 6.41 6.42/6.43 6.52 6.59

6.9 $y=A\cos[\omega(\frac{x}{u}+t)+\varphi]=A\sin[-\omega(\frac{x}{u}+t)+\frac{\pi}{2}-\varphi]$

振幅 $A=6cm$, $\frac{\omega}{u}=0.02\pi$, $\varphi=0$, $\omega=4\pi$ $\Rightarrow$ 波速 $u=200cm/s$

周期 $T=\frac{2\pi}{\omega}=0.5s$ 频率 $f=2Hz$ 波长 $\lambda=uT=1m$

波传播方向为 x 轴负向 $v_{max}=\omega A=24\pi cm/s$

6.12 $y=A\cos[\omega(t-\frac{x}{u})+\varphi]$

$f=0.5Hz \Rightarrow \omega=\pi rad/s$, $A=1\times10^{-2}m$, $\lambda=4m \Rightarrow u=\frac{\lambda}{T}=2m/s$

$t=\frac{1}{3}\Rightarrow y=1\times10^{-2}\cos(\frac{1}{3}\pi-\frac{\pi}{2}x+\varphi)$

$\Rightarrow \varphi=\frac{\pi}{3}$

$\therefore y=1\times10^{-2}\cos[\pi t-\frac{\pi}{2}x+\frac{\pi}{3}]$

$x=0$ 时, $y=1\times10^{-2}\cos(\pi t+\frac{\pi}{3})$

距离为 $\frac{2.5}{3}m$, $x=\frac{2}{3}m$ 时, $y=1\times10^{-2}\cos(\pi t-\frac{5}{6}\pi)$

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6.21 $u=\sqrt{\frac{F}{\mu}}=12m/s$

$A=0.04\sqrt{2}m$, $\lambda=0.4m$, $\omega=60\pi rad/s$

$V_{max}=\omega A=2.4\sqrt{2}\pi m/s$

$y=A\cos[\omega(t+\frac{x}{u})+\varphi]$

$\Rightarrow y=0.04\sqrt{2}\cos[60\pi t+5\pi x+\varphi]\Rightarrow\varphi=-\frac{\pi}{4}$

$\Rightarrow y=0.04\sqrt{2}\cos(60\pi t+5\pi x-\frac{\pi}{4})$

6.27 $\bar{w}=\frac{1}{2}\rho A^2\omega^2=16\pi^2 J/m^3$

$I=\bar{w}u=1.6\times10^4\pi^2 W/m^2$

$E=ISt=384\pi^2 J$

6.34 $y_1=A_1\cos[\omega(t+\frac{r_1}{u})+\varphi_1]$

$y_2=A_2\cos[\omega(t-\frac{r_2}{u})+\varphi_2]$

$\Rightarrow A=\sqrt{A_1^2+A_2^2+2A_1A_2\cos\Delta\varphi}$

$\Delta\varphi=\varphi_2-\varphi_1-\frac{2\pi}{\lambda}(r_2-r_1)$

$\varphi_2-\varphi_1=\frac{\pi}{2}$

对S1外侧点，$r_2-r_1=\frac{\lambda}{4}$，$\Delta\varphi=0$ $A=2A_0$，$I=4I_0$

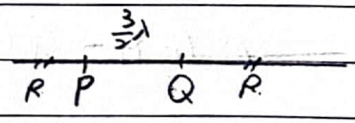
对S2外侧点，$r_2-r_1=\frac{\lambda}{4}$，$\Delta\varphi=\pi$ $A=0$，$I=0$

6.36 $y_P=A_0\cos[\omega(t-\frac{r_1}{u})]$

$y_Q=A_0\cos[\omega(t-\frac{r_2}{u})]$

$\Rightarrow A=\sqrt{2A_0^2+2A_0^2\cos\Delta\varphi}$

$\Delta\varphi=-\frac{2\pi}{\lambda}(r_2-r_1)$



$|r_2-r_1|=\frac{3}{2}\lambda$ 时，$\Delta\varphi=\pm\pi$，$A=0$

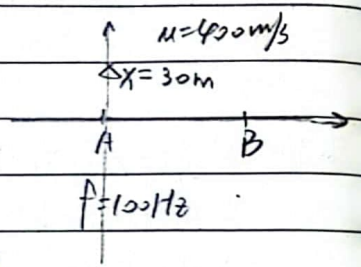
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6.41 $y_2 = A\cos(\omega t + kx - 6k + \pi)$

$y = y_1 + y_2 = A[\cos(\omega t - kx) + \cos(\omega t + kx - 6k + \pi)]$

$= 2A\cos(\omega t - 3k + \frac{\pi}{2})\cos(-kx + 3k - \frac{\pi}{2})$

6.42 $\lambda = \frac{u}{f} = 4m$



$|P_B - P_A| = 2\pi$，不妨令 $P_B - P_A = 2\pi$

$\Delta\varphi = P_B - P_A - \frac{2\pi}{\lambda}(r_B - r_A) = 2\pi + 2k\pi$

$\Rightarrow r_B - r_A = -k\lambda \ (k \in Z)$

$\Delta x - x - x = -k\lambda \Rightarrow x = \frac{\Delta x + k\lambda}{2}$

$\Rightarrow x = (15 + 2k)m \quad k \in [-7, 7]$ 且 $k \in Z$

6.45 $y_0 = A\cos(2\pi\frac{t}{T} + \frac{\pi}{2})$

$\therefore y_2 = A\cos[2\pi(\frac{t}{T} - \frac{x}{\lambda}) + \frac{\pi}{2}]$

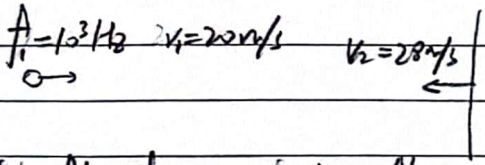
$y = y_1 + y_2 = A\{\cos[2\pi(\frac{t}{T} - \frac{x}{\lambda}) + \frac{\pi}{2}] + \cos[2\pi(\frac{t}{T} + \frac{x}{\lambda}) + \frac{\pi}{2}]\}$

$= 2A\cos(2\pi\frac{t}{T} + \frac{\pi}{2})\cos 2\pi\frac{x}{\lambda}$

$2\pi\frac{x}{\lambda} = k\pi$，$x = \frac{k\lambda}{2}$ 为波腹

$2\pi\frac{x}{\lambda} = \frac{\pi}{2} + k\pi$，$x = (k + \frac{1}{2})\lambda$ 为波节

6.52



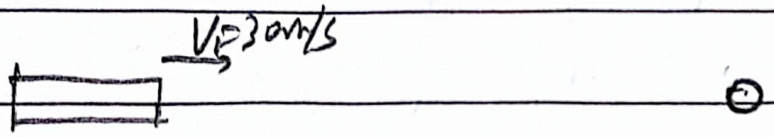
$f_2 = \frac{u}{u - v_1} f_1$ $\quad f_2\lambda = \frac{u}{f_2} = 0.32m$

$f_3 = \frac{v_2 + u}{u - v_1} f_1$ $\quad f_3 = 1.15 \times 10^3 Hz$ $\quad N = 1.15 \times 10^3$

$f_4 = \frac{u_0}{u - v_2} f_3$ $\quad f_4 = 1.25 \times 10^3 Hz$ $\quad \lambda = \frac{u}{f_4} = 0.27m$

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6.59 $f = 440\,Hz$ $V_0 = 20\,m/s$ $u = 340\,m/s$



$V_1' = V_1 = V_0 = 40\,m/s$

$V_2' = V_0 = 20\,m/s$

$$f' = \frac{u + V_2'}{u - V_1'} f = 480\,Hz$$

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